

BGlab1 (part1+part2)

姓 名 杨力泉

学 号 231220069

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一、实验环境

- 操作系统：Ubuntu 22.04 (WSL2)
- JDK 版本：OpenJDK 1.8.0_281
- Hadoop 版本：3.2.1
- 硬件配置：vCPU2核，内存4GB，硬盘30G

二、实验内容与步骤

1. 环境准备与配置文件修改

- 安装 SSH、JDK 8，配置环境变量（详见附录）。
- 生成 SSH 密钥并配置免密登录。
- `core-site.xml`、`hdfs-site.xml`、`yarn-site.xml`、`mapred-site.xml` 按手册配置。

2. 启动 Hadoop 集群

- 启动 HDFS 和 YARN 进程(截图如下):

```
hk416@hk416-virtual-machine:~/hadoop_installs/hadoop-3.2.1$ sbin/start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [hk416-virtual-machine]
hk416@hk416-virtual-machine:~/hadoop_installs/hadoop-3.2.1$ sbin/start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

```
hk416@hk416-virtual-machine:~/hadoop_installs/hadoop-3.2.1$ jps
3218 DataNode
3092 NameNode
3719 ResourceManager
3451 SecondaryNameNode
4219 Jps
3868 NodeManager
```

- HDFS 命令行基本操作

```
hk416@hk416-virtual-machine:/home$ hdfs dfs -mkdir -p /user/hk416/test-in
hk416@hk416-virtual-machine:/home$ hdfs dfs -put ~/test_data/*.txt /user/hk416/test-in/
2026-03-24 14:23:11,773 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
2026-03-24 14:23:12,538 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
hk416@hk416-virtual-machine:/home$ hdfs dfs -ls /user/hk416/test-in/
```

- HDFS和YARN的web界面

Overview 'localhost:9000' (active)

Started:	Tue Mar 17 20:18:07 +0800 2026
Version:	3.2.1, rb3cbbb467e22ea829b3808f4b7b01d07e0bf3842
Compiled:	Tue Sep 10 23:56:00 +0800 2019 by rohithsharmaks from branch-3.2.1
Cluster ID:	CID-01e27a52-b033-48d3-9f50-e30adb1a8b9f
Block Pool ID:	BP-649459645-127.0.1.1-1773749799367

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localhost:8088/cluster

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All Applications

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory
0	0	0	0	0	0 B	8 GB	0 B

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes
1	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>

Show 20 entries

ID	User	Name	Application Type	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCores	Allocated Memory MB	Rese Cf VCc
No data available in table														

Showing 0 to 0 of 0 entries

- 通过 jps 命令和 Web 界面的检查，所有 Hadoop 进程均正常运行，表明伪分布式环境搭建成功。HDFS 命令行操作结果与预期一致，文件的上传、下载、查看、删除功能正常，证明 HDFS 客户端与 NameNode 和 DataNode 通信正常。

3.执行 Hadoop 官方示例程序 WordCount

- 本地创建文件之后，在 HDFS 中创建输入目录并上传文件，运行程序，执行过程中，终端会输出 MapReduce 作业的进度信息，包括 Map 和 Reduce 任务的启动、执行、完成情况。最后会显示 Job completed successfully 的提示。

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```
hk416@hk416-virtual-machine:~/hadoop_installs/hadoop-3.2.1$ bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-3.2.1.jar wordcount /user/hk416/test-in /user/hk416/test-out
2026-03-17 20:36:11,668 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-17 20:36:12,185 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hk416/.staging/job_1773750679233_0001
2026-03-17 20:36:12,306 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
2026-03-17 20:36:12,457 INFO input.FileInputFormat: Total input files to process : 2
2026-03-17 20:36:12,519 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
2026-03-17 20:36:12,560 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
2026-03-17 20:36:12,983 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-17 20:36:13,197 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
```

- 最后的结果如下图

```
hk416@hk416-virtual-machine:~/hadoop_installs/hadoop-3.2.1$ bin/hdfs dfs -cat /user/hk416/test-out/part-r-00000
2026-03-17 20:36:38,868 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
goodbye 1
hadoop 2
hello 2
world 1
```